Wi-Fi 5 / 6 / 6E Protocol, Operation and Deployment

Course objectives:

Wi-Fi 5 (802.11ac) and Wi-Fi 6 (802.11ax) are the latest generations of wireless networking standards. They offer significant improvements in speed, capacity, and efficiency compared to their predecessors. Operating in the license-exempt frequency band, Wi-Fi is one of the greatest success stories in modern technology era.

In this training course we'll explore the key features of each of these Wi-Fi generations and discuss the most important protocol functions, required to understand the aspects of a successful Wi-Fi network deployment in the most common scenarios: residential Wi-Fi deployment, enterprise indoor deployment and outdoor Wi-Fi deployment.

The participant will learn about network topologies, frequency bands, Wi-Fi protocol operation and security features.

Duration Instructional methods:

Option 1: On-Site Training: 2 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 2 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 4 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

The course is addressed to technical and non-technical staff from the technology, marketing and management field, regulators and government officials, executives, requiring an overview about the latest Wi-Fi technologies, the protocols behind the operation and possible deployment scenarios.

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Course Contents:

- Introduction to Wi-Fi
  - Background and drivers for Wi-Fi standards
  - Role of 3EEE and Wi-Fi Alliance
  - Frequency spectrum aspects
  - Wi-Fi in WLAN environment and ad-hoc operation

- General Wi-Fi Overview: MAC and PHY layers
  - Wi-Fi Network Architecture
  - MAC Frame Formats and Types
  - Medium Access
  - AP Discovery
  - Connection establishment and termination
  - Quality of Service
  - STA Power Save Methods
  - STA Roaming
  - AP Channel Switching
  - PHY layer description for Wi-Fi 4 (802.11n) and Wi-Fi5 (802.11ac)

- Wi-Fi6 (802.11ax)
  - The HE PPDU Formats
  - Single-User Operation
  - Multi-User Operation
  - Target Wake Time
  - BSS Coloring aspects
  - Spatial Reuse
  - Enhanced Multi-BSSID Advertisement
  - Operation Mode Indication
  - HE Capabilities Advertisement
  - SU, MU-MIMO and MU-OFDMA use cases

- Wi-Fi Security
  - WEP and its vulnerabilities
  - WPA/WPA2 operation aspects
  - Key Re-Installation Attack
  - WPA3 operation
  - Multi-Channel Man-In-The-Middle Attack
  - Fragmentation and Aggregation Attacks
  - Rogue AP Detection

- Wi-Fi Implementation
  - The Hardware
  - The Software and Algorithms
  - Regulatory Requirements
  - Wi-Fi Certification

- Wi-Fi 6E
  - Operation in the 6GHz Band
  - AP Discovery
  - Enhanced Multi-BSSID Advertisement (EMA) in 6GHz
  - Beacon Advertisement and Channel Rates in 6GHz
  - 6 GHz Regulatory Requirements
  - Wi-Fi 6E Implementation Challenges
  - 160MHz Multiple AP Deployment in 6GHz

- Wi-Fi Deployment Aspects
  - Residential Wi-Fi Deployment
  - Enterprise Indoor Wi-Fi Deployment
  - Outdoor Wi-Fi Deployment
  - AP Placement
  - Channel Bandwidth Selection
  - Estimation of Number of AP's
  - Mitigation of Inter-AP Interference
  - Upgrading existing deployments
  - Post-deployment Site Survey

Pre-requisites:

Basic knowledge about IP networks and experience with IT infrastructure is beneficial.

Course Code: WiFi6